

$$\begin{aligned}
 5: 1 \times 3^1 + 2 \times 3^0 &\rightarrow 012 & 6: 2 \times 3^1 + 0 \times 3^0 &\rightarrow 020 & 7: 2 \times 3^1 + 1 \times 3^0 &\rightarrow 021 \\
 8: 2 \times 3^1 + 2 \times 3^0 &\rightarrow 022 & 9: 1 \times 3^2 + 0 \times 3^1 + 0 \times 3^0 &\rightarrow 100 \\
 10: 1 \times 3^2 + 0 \times 3^1 + 1 \times 3^0 &\rightarrow 101 & 11: 1 \times 3^2 + 0 \times 3^1 + 2 \times 3^0 &\rightarrow 102 \\
 12: 1 \times 3^2 + 1 \times 3^1 + 0 \times 3^0 &\rightarrow 110 & 13: 1 \times 3^2 + 1 \times 3^1 + 1 \times 3^0 &\rightarrow 111 \\
 14: 1 \times 3^2 + 2 \times 3^1 + 0 \times 3^0 &\rightarrow 120 & 15: 1 \times 3^2 + 2 \times 3^1 + 1 \times 3^0 &\rightarrow 121 \\
 16: 1 \times 3^2 + 2 \times 3^1 + 2 \times 3^0 &\rightarrow 122 & 17: 2 \times 3^2 + 0 \times 3^1 + 0 \times 3^0 &\rightarrow 200 \\
 18: 2 \times 3^2 + 0 \times 3^1 + 1 \times 3^0 &\rightarrow 201 & 19: 2 \times 3^2 + 1 \times 3^1 + 2 \times 3^0 &\rightarrow 212 \square
 \end{aligned}$$

1-2: Add and multiply the following numbers in the given base without converting to decimal:

(a) $(1230)_4$ and $(23)_4$

$$\begin{array}{r}
 1230 \\
 + 23 \\
 \hline
 1313
 \end{array}$$

$$\begin{array}{r}
 1230 \\
 \times 23 \\
 \hline
 1010 \\
 31200 \\
 \hline
 32210
 \end{array}$$

1-3: Convert the decimal number 250.5 to base 3, base 4, base 7, base 8 and base 16.

$$250 + 0.5 = 250.5$$

$$(0.5) \times 3 = 1.5 = 1 + 0.5$$

$$0.5 \times 3 = 1 + 0.5 \dots$$

$$(250.5)_3 = 100021.11\dots$$

$$\begin{array}{r}
 3 \overline{) 250} \\
 \underline{83} \\
 27 \\
 \underline{9} \\
 3 \\
 \underline{3} \\
 1 \\
 \underline{0} \\
 0
 \end{array}$$

$$(250)_3 = 100021$$

$$\begin{array}{r}
 4 \overline{) 250} \\
 \underline{62} \\
 15 \\
 \underline{12} \\
 3 \\
 \underline{0} \\
 0
 \end{array}$$

$$(250)_4 = 3322$$

$$(250.5)_4 = 3322.2$$

$$\begin{array}{r}
 7 \overline{) 250} \\
 \underline{35} \\
 5 \\
 \underline{5} \\
 0
 \end{array}$$

$$0.5 \times 7 = 3.5 = 3 + 0.5$$

$$0.5 \times 7 = 3.5 = 3 + 0.5 \dots$$

$$(250.5)_7 = 505.333\dots$$

$$\begin{array}{r}
 8 \overline{) 250} \\
 \underline{32} \\
 4 \\
 \underline{0} \\
 0
 \end{array}$$

$$(0.5) \times 8 = 4.0 = 4 + 0.0$$

$$\therefore (250.5)_8 = 413.4$$

$$(0.5) \times 16 = 8 + 0.0$$

$$\begin{array}{r}
 16 \overline{) 250} \\
 \underline{16} \\
 0
 \end{array}$$

$$\therefore (250.5)_{16} = FA.8$$